INSIGHT | ACTION | CONTROL

ISPE UK · TAKE-AWAY CHECKLIST

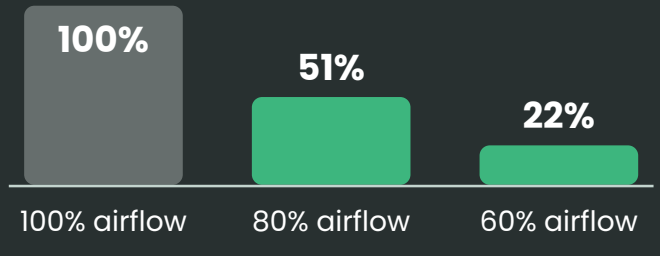
2-MINUTE SELF-CHECK

Is your cleanroom wasting energy?

Tick every statement that's true of your site. Each tick points to a saving you can usually make **while staying in grade**.

WHY AIRFLOW MATTERS

-20% airflow
≈ -49% fan power



Fan power ≈ airflow³ (fan affinity laws, ideal case). Small airflow cuts, big energy savings.

Operational efficiency

- [ ] Air change rates haven't been reviewed since commissioning.
- [ ] Fans run at full speed overnight and at weekends.

Sustainability & carbon

- [ ] We don't know what share of site energy our cleanrooms use.
- [ ] Our carbon targets leave GMP areas out of scope.

Digitalisation & data

- [ ] Particle data is logged for compliance, never used to control airflow.
- [ ] We can't see energy use per room or suite.

Compliance

- [ ] Energy projects stall because nobody's sure how QA will react.
- [ ] Energy measures aren't part of our Contamination Control Strategy.

Future-ready design

- [ ] Our next build or refit is designed around fixed, worst-case airflow.
- [ ] Our BMS can't adjust airflow to live room conditions.

COUNT YOUR TICKS

0–2
Well tuned
Prove it: benchmark your site against your peers.

3–5
Quick wins likely
See where to look first, overleaf.

6–10
Big opportunity
Worth a conversation. Scan the QR code overleaf.

Turn over: two places to look first →

Guidance only, not a site assessment.

TICKED A FEW?

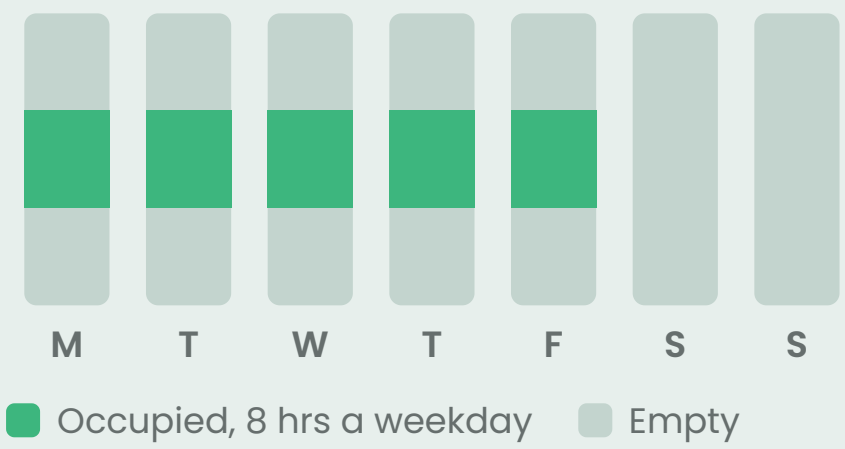
Two places to look first

1 Empty rooms, full-speed fans

If you ticked: "Fans run at full speed overnight and at weekends."

76%

of the week, a single-shift cleanroom is empty



Look at: a validated setback mode that steps fans down when rooms are empty.

2 Airflow set by design, not data

If you ticked: "Particle data is logged for compliance, never used to control airflow."

54%

of site energy went on air handling at a pharma campus we studied



We identified a **28% saving** on that air handling energy, with no added GMP risk.

Look at: how far below grade limits your particle counts sit.

How many did you tick?
Let's talk them through.

Scan to talk through your ticks with an EECO2 engineer, and find out where your savings are.

SCAN TO TALK TO US

eeco2.com/contact

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References: ISO 14644-16:2019 Cleanrooms and associated controlled environments – Energy efficiency; EU GMP Annex 1 (2022) Manufacture of Sterile Medicinal Products; fan affinity laws (ideal case). Guidance only, not a site assessment.